

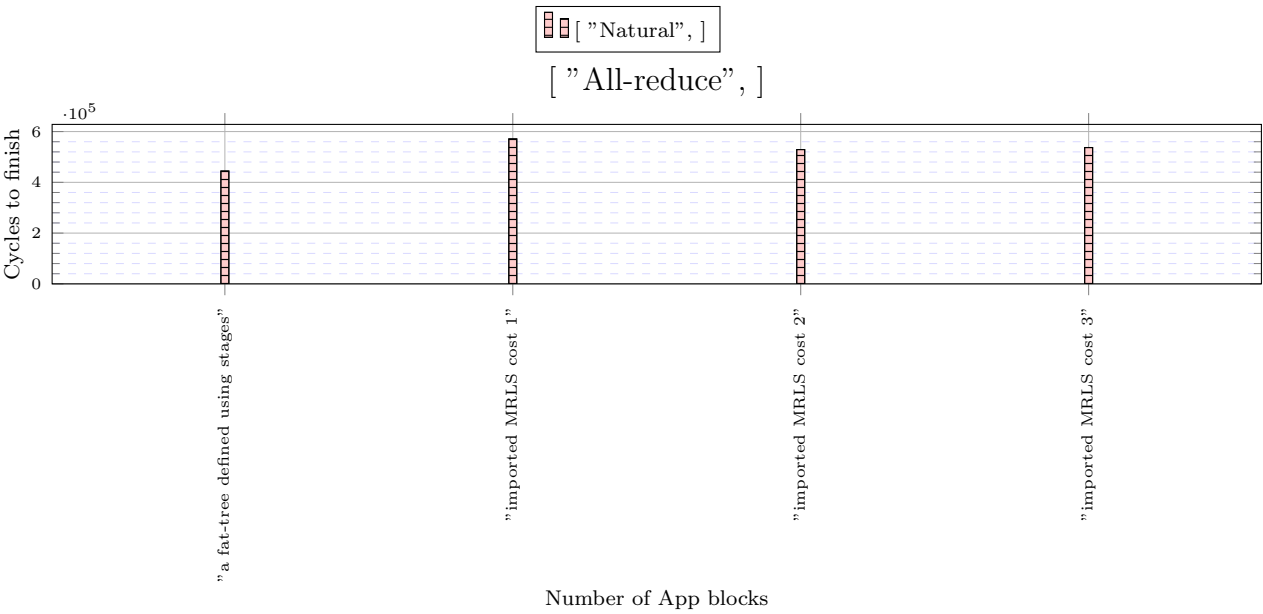


Figure 1: X: ["All-reduce",]

- heads/alex-dev-a8bf2d5ee67d9d0c7dceda54f01ab7cede4ab269(0.6.3)
- heads/alex-dev-8e2a39f34b812569e780c03ed6a7e158aa4b09bc(0.6.3)

```

Configuration{
random_seed: ![ 42, 935, 128643 ],
warmup: 9999999999999,
measured: 9999999999999,
general_frequency_divisor: 2,
topology: topo![
  MultiStage{
    stages: [
      Fat{ bottom_factor: 18, top_factor: 18 },
      Fat{ bottom_factor: 18, top_factor: 18 },
      Fat{ bottom_factor: 18, top_factor: 18 }],
    servers_per_leaf: 18,
    legend_name: "a fat-tree defined using stages"},
  MultiStage{
    stages: [
      ExplicitStageFile{ filename: "../MRLS_5832_18_2916_36" }],
    servers_per_leaf: 18,
    legend_name: "imported MRLS cost 1"},
  MultiStage{
    stages: [
      ExplicitStageFile{ filename: "../MRLS_8748_24_5832_36" }],
    servers_per_leaf: 12,
    legend_name: "imported MRLS cost 2"},
  MultiStage{
    stages: [
      ExplicitStageFile{ filename: "../MRLS_11664_27_8748_36" }],
    servers_per_leaf: 9,
    legend_name: "imported MRLS cost 3"}],
traffic: ![
  TrafficMap{
    tasks: 104976,
    map: ![
      Composition{
        patterns: [
          Identity{ legend_name: "Natural" },
          CartesianEmbedding{
            source_sides: [ 65536 ],
            destination_sides: [ 104976 ]}],
        middle_sizes: [ 65536, 104976 ],
        legend_name: "Natural"}],
        application: ![
          AllReduce{ tasks: 65536, data_size: 65536 },
          legend_name: "All-reduce"}],
maximum_packet_size: 16,
router: InputOutput{
  virtual_channels: topo![ 3, 4, 4, 4 ],
  virtual_channel_policies: topo![
    [
      OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },
      OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
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      EnforceFlowControl,
      Random],
    [
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          ArgumentVC{
            allowed: [ 0 ]},
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          ArgumentVC{
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          ArgumentVC{
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          ArgumentVC{
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          OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },
          OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: false },
          LowestLabel,
          EnforceFlowControl,
          Random],
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              ArgumentVC{
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              ArgumentVC{
                allowed: [ 2 ]},
              ArgumentVC{
                allowed: [ 2 ]},
              ArgumentVC{
                allowed: [ 3 ]},
              ArgumentVC{
                allowed: [ 3 ]}],
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```

```

    ArgumentVC{
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    ArgumentVC{
      allowed: [ 3 ]}],
  VecLabel{
    label_vector: [ 0, 64, 80 ]},
  OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },
  OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: false },
  LowestLabel,
  EnforceFlowControl,
  Random],
[
  MapHop{
    hop_to_policy: [
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      ArgumentVC{
        allowed: [ 0 ]},
      ArgumentVC{
        allowed: [ 1 ]},
      ArgumentVC{
        allowed: [ 1 ]},
      ArgumentVC{
        allowed: [ 2 ]},
      ArgumentVC{
        allowed: [ 2 ]},
      ArgumentVC{
        allowed: [ 3 ]},
      ArgumentVC{
        allowed: [ 3 ]}],
    VecLabel{
      label_vector: [ 0, 64, 80 ]},
    OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },
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use_neighbour_space: true, aggregate: false },
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    EnforceFlowControl,
    Random]],
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  buffer_size: 128,
  bubble: false,
  flit_size: 16,
  intransit_priority: false,
  allow_request_busy_port: true,
  output_buffer_size: 64,
  crossbar_frequency_divisor: 1,
  crossbar_delay: 2},
routing: topo![
  UpDown{ legend_name: "minimal routing" },
  Polarized{ include_labels: true, legend_name: "Polarized original" },
  Polarized{ include_labels: true, legend_name: "Polarized original" },
  Polarized{ include_labels: true, legend_name: "Polarized original" }],
link_classes: [
  LinkClass{ delay: 2 },
  LinkClass{ delay: 2 },
  LinkClass{ delay: 2 },
  LinkClass{ delay: 2 }],
launch_configurations: [
  Slurm{ job_pack_size: 1, time: "35-00:00:00", mem: "70G" }]}

```